

RACHANA VARSHA VELLAICHAMY SRISELVAM

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Experience

ShineOnYouth(Non-Profit Organization)

Oct 2024 – Present

ML Engineer

Frisco, Texas

- Spearheaded the development of a comprehensive website for Shine On Youth, improving access to self-esteem resources for over 1,000 teens, and achieving a 25% improvement in user interaction metrics.
- Developed VoiceVibe Insight, a real-time sentiment analysis web app using Flask and the Web Speech API, enabling live audio transcription and sentiment scoring.
- Engineered custom APIs and integrated real-time voice processing pipelines to ensure seamless user interaction and low-latency feedback.

OnTrac | Python, SQL, TensorFlow, GCP, Data Pipeline Development

July 2024 – Oct 2024

Data Engineering Intern

Atlanta, Georgia

- Independently built a comprehensive data dictionary, significantly improving data consistency and accessibility across departments. Developed and deployed **Text-to-SQL** pipelines on Google Cloud Platform (GCP), which enabled seamless natural language processing (NLP) and SQL query generation. Mastered **prompt engineering** to dynamically translate user questions into SQL queries, incorporating text embeddings and data dictionary references for context-specific query generation
- Faced with the challenge of minimizing the manual effort involved in SQL query generation, I designed and directed a **conversational AI chatbot** using **Vertex AI** and **NLP** models. This innovative chatbot successfully translated user queries into optimized SQL statements, drastically reducing the need for manual SQL query writing by the data analytics team and defined data quality monitoring systems using Apache Spark for large-scale data processing, ensuring data integrity across multiple sources.
- Managed the development and maintenance of end-to-end **ETL** (Extract, Transform, Load) pipelines using Python, TensorFlow, and Keras for natural language processing tasks to streamline data movement from the landing zone to the hub. This effort ensured high data availability and quality, reduced data processing time by 15%, and minimized disruptions in analytics and reporting processes.

DTCC | Python, SQL, Apache Airflow, GCP (BigQuery, Cloud Storage), dbt

Mar 2021 – June 2021

Data Engineering Intern

Chennai, TamilNadu

- Built a data ingestion pipeline to load employee onboarding data from GCP Cloud Storage into BigQuery using Python and dbt, enabling scalable and queryable analytics with minimal latency.
- Designed modular dbt models to standardize employee records—parsing department hierarchy, normalizing roles/titles, and flagging incomplete onboarding steps, enabling HR teams to run real-time queries for status tracking and compliance.

Projects

SKILL JAM CONNECT | AWS, Java, REST APIs

May 2024

- Developed Skill Jam Connect, a mobile learning application designed for consistent performance across various platforms, integrating backend services to ensure smooth and responsive user experiences.
- Developed Skill Jam Connect, a mobile learning application designed for consistent performance across platforms, using Java for backend logic and API development and integrated backend services using Java and Spring Boot, ensuring smooth and responsive user experiences.
- Deployed the app as a cloud-native solution on AWS, incorporating RESTful APIs to support real-time content delivery and user authentication and built internal tools for monitoring and debugging REST endpoints, enhancing system stability and reducing response errors during peak usage.

IMAGE RENDERING INTELLIGENT SYSTEM(IRIS) | OpenCV, TensorFlow, Pytorch, Golang

May 2022

- Innovated a software solution to eliminate unwanted obstructions from video footage, offering both manual and automated modes of obstruction removal. Composed frame-by-frame image processing techniques using OpenCV to analyze and enhance video frames, significantly improving the accuracy of obstruction detection and removal.
- Implemented an algorithm that leverages **motion vector analysis** to differentiate between static backgrounds and dynamic obstructions, reducing processing time by 30% and enhancing overall video clarity.
- Built a distributed video processing system using Golang and AWS Lambda for handling concurrent processing of multiple video streams. Streamlined machine learning models using TensorFlow and PyTorch for content recommendation system.

Education

University of Texas at Arlington

Aug 2022 – May 2024

Masters in Computer Science (Specialization: Databases and Machine Learning)

Arlington, Texas

Loyola ICAM College of Engineering and Technology

Aug 2018 – June 2022

Bachelor of Engineering in Computer Science

Chennai, TamilNadu

Technical Skills

Certification: AWS Certified AI Practitioner

Languages: Java, Python, SQL, HTML, CSS, React.js, Golang, .Net

Developer Tools: Jupyter Notebook, VS Code, Eclipse, Google Cloud Platform, Android Studio, WordPress, GitHub

Databases: SAS SQL, MySql, Oracle, SSMS, Big Data: Spark, Hadoop/MapReduce

Technologies/Frameworks: Apache Spark, Microservices Architecture, REST APIs, AWS(EC2, S3, Lambda, SQS, Step Functions), Azure DevOps, GCP, ETL Processes, Data Warehousing, Prompt Engineering, Libraries(Pandas, NumPy, SciPy, Scikit-learn)